From Fields to Completeness: A Gentle Introduction to the Real Numbers for Business and Economics Students

Wesley's Note on the Completeness of Real Numbers
(實數的完備性)

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1 Why Do We Start with Fields When Discussing the Completeness of Real Numbers?

Before studying the completeness of the real numbers, many textbooks in real analysis first introduce the concept of a *field*. At first glance, this definition may look abstract and formal, especially for students in business, economics, or finance. However, the idea behind a field is actually quite simple: a field is a number system in which the four basic arithmetic operations—addition, subtraction, multiplication, and division—can be performed in a stable and consistent way.

We first present the standard textbook definition.[1]

Definition 1 (Field) *A set F is called a* field *if it has two operations defined on it, addition $x + y$ and multiplication xy, and if it satisfies the following axioms:*

(A1) If $x \in F$ and $y \in F$, then $x + y \in F$.

(A2) ***Commutativity of addition:***

$$x + y = y + x$$

for all $x, y \in F$.

(A3) ***Associativity of addition:***

$$(x + y) + z = x + (y + z)$$

for all $x, y, z \in F$.

(A4) There exists an element $0 \in F$ such that

$$0 + x = x$$

for all $x \in F$.

(A5) For every element $x \in F$, there exists an element $-x \in F$ such that

$$x + (-x) = 0.$$

(M1) If $x \in F$ and $y \in F$, then $xy \in F$.

[1] See Jiří Lebl, *Basic Analysis I: Introduction to Real Analysis, Volume I*, Section 1.1. The textbook is available as a free online textbook on the author's website.

(M2) ***Commutativity of multiplication:***

$$xy = yx$$

for all $x, y \in F$.

(M3) ***Associativity of multiplication:***

$$(xy)z = x(yz)$$

for all $x, y, z \in F$.

(M4) There exists an element $1 \in F$*, with* $1 \neq 0$*, such that*

$$1x = x$$

for all $x \in F$.

(M5) For every $x \in F$ *such that* $x \neq 0$*, there exists an element* $1/x \in F$ *such that*

$$x(1/x) = 1.$$

(D) ***Distributive law:***

$$x(y + z) = xy + xz$$

for all $x, y, z \in F$.

Although the definition contains many axioms, its main message is straightforward. A field is a set in which addition, subtraction, multiplication, and division behave in the usual way. In particular, the results of these operations remain inside the same set.

For example, the set of rational numbers,

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z},\ b \neq 0 \right\},$$

is a field. If we take two rational numbers such as $\frac{1}{2}$ and $\frac{1}{3}$, then

$$\frac{1}{2} + \frac{1}{3} = \frac{5}{6},$$

and $\frac{5}{6}$ is still a rational number. Similarly,

$$\frac{1}{2} \times \frac{3}{4} = \frac{3}{8},$$

and $\frac{3}{8}$ is still a rational number. Even division works, as long as we are not dividing by zero:

$$\frac{1}{2} \div \frac{3}{4} = \frac{1}{2} \times \frac{4}{3} = \frac{2}{3}.$$

Thus, rational numbers are closed under the basic arithmetic operations required in a field.

This is very different from the set of integers,

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}.$$

The integers are closed under addition, subtraction, and multiplication. For example,

$$2 + 3 = 5 \in \mathbb{Z},$$

and

$$2 \times 3 = 6 \in \mathbb{Z}.$$

However, the integers are not closed under division. For example,

$$1 \div 2 = \frac{1}{2},$$

but

$$\frac{1}{2} \notin \mathbb{Z}.$$

Therefore, $\mathbb{Z}$ is not a field.

More precisely, the property that fails is the existence of multiplicative inverses. In a field, every nonzero element x must have a multiplicative inverse $1/x$ such that

$$x \cdot \frac{1}{x} = 1.$$

In $\mathbb{Q}$, the number 2 has the multiplicative inverse $\frac{1}{2}$, because

$$2 \cdot \frac{1}{2} = 1.$$

But in $\mathbb{Z}$, the element $\frac{1}{2}$ is not available. Hence, 2 does not have a multiplicative inverse inside $\mathbb{Z}$. This is why the integers are not a field.

From the perspective of business, economics, and finance, the importance of a field is quite intuitive. Many basic formulas require not only addition and multiplication, but also division. For instance, the standard present-value formula is

$$PV = \frac{CF}{1+r},$$

where PV is present value, CF is future cash flow, and r is the discount rate. This formula implicitly requires that $1 + r$ has a multiplicative inverse. Indeed, we can rewrite it as

$$PV = CF \cdot \frac{1}{1+r}.$$

Therefore, whenever we perform discounting, valuation, return calculation, portfolio optimization, or utility maximization, we are implicitly working in a number system where these algebraic operations are valid. This is one reason why field structure is a natural starting point for mathematical economics and finance.

However, being a field only guarantees good algebraic behavior. It tells us that addition, subtraction, multiplication, and division can be performed consistently. It does not yet tell us whether we can compare numbers, nor does it tell us whether limits exist. For these purposes, we need additional structure.

The next step is the concept of an *ordered field*. An ordered field is a field in which we can also compare elements. That is, for two elements x and y, we can meaningfully discuss whether

$$x < y, \qquad x = y, \qquad \text{or} \qquad x > y.$$

Once we have an order, we can define concepts such as upper bounds, lower bounds, increasing sequences, decreasing sequences, maximum points, and minimum points. These concepts are essential for discussing limits, continuity, optimization, and completeness.

The rational numbers $\mathbb{Q}$ are not only a field; they are also an ordered field. We can add, subtract, multiply, and divide rational numbers, and we can also compare their sizes. For example,

$$\frac{1}{2} < \frac{2}{3}.$$

At this stage, $\mathbb{Q}$ seems to be a very good number system. It supports arithmetic operations and order comparisons. Nevertheless, it has a fundamental weakness: it is not complete.

The incompleteness of $\mathbb{Q}$ can be understood intuitively as the existence of holes on the rational number line. The classical example is $\sqrt{2}$. We know that

$$\sqrt{2} \approx 1.41421356\ldots,$$

so $\sqrt{2}$ lies between 1 and 2. However, $\sqrt{2}$ is not a rational number:

$$\sqrt{2} \notin \mathbb{Q}.$$

This means that even though rational numbers are very dense on the number line, they do not fill the entire line. There are still missing points.

This issue becomes especially clear when we consider sequences. For example, take the sequence

$$1, \quad 1.4, \quad 1.41, \quad 1.414, \quad 1.4142, \quad \ldots$$

Every term in this sequence is a rational number, because each term can be written as a finite decimal. Intuitively, this sequence gets closer and closer to $\sqrt{2}$. However, if we restrict ourselves to the rational numbers, the value that the sequence approaches is not contained in the system. The sequence is trying to converge to something, but that "something" is missing from $\mathbb{Q}$.

This is the sense in which the rational numbers are not complete.

The completeness of the real numbers $\mathbb{R}$ solves this problem. The real numbers form not only a field, and not only an ordered field, but a *complete ordered field*. Intuitively, completeness means that the real number line has no holes. Whenever a sequence should converge, its limit exists as a real number.

A standard way to express completeness is through the least-upper-bound property. It states that every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound, also called the supremum.

For example, consider the set

$$A = \{x \in \mathbb{Q} : x^2 < 2\}.$$

This set contains all rational numbers whose squares are less than 2. The set is bounded above. For instance, 2 is an upper bound. However, in the rational number system $\mathbb{Q}$, this set has no least upper bound. The number that should serve as the least upper bound is

$$\sqrt{2},$$

but $\sqrt{2}$ is not rational. Hence, the least upper bound is missing from $\mathbb{Q}$.

In contrast, in the real number system $\mathbb{R}$, the number $\sqrt{2}$ exists. Therefore, the same set has a least upper bound in $\mathbb{R}$. This illustrates the essential difference between $\mathbb{Q}$ and $\mathbb{R}$. Both systems allow arithmetic operations and order comparisons, but only $\mathbb{R}$ is complete.

This explains why textbooks introduce fields before discussing the completeness of the real numbers. The logical structure is as follows:

$$\text{Field} \quad \Rightarrow \quad \text{Ordered Field} \quad \Rightarrow \quad \text{Complete Ordered Field}.$$

A field answers the question: can we perform addition, subtraction, multiplication, and division consistently? An ordered field answers the question: can we compare numbers? A complete ordered field answers the question: when we take limits or use approximation, does the limiting value remain inside the number system?

In this hierarchy, the rational numbers $\mathbb{Q}$ are a field and an ordered field, but they are not complete. The real numbers $\mathbb{R}$, by contrast, form a complete ordered field. This is why $\mathbb{R}$ is the natural foundation for calculus, real analysis, optimization, probability theory, econometrics, and financial mathematics.

For students in economics and finance, this point is more than a technical detail. In economic and financial models, variables such as prices, returns, interest rates, consumption, investment, utility, probabilities, and estimators are typically treated as real numbers. This is not merely a matter of convenience. The real number system has the algebraic structure needed for calculation, the order structure needed for comparison and optimization, and the completeness structure needed for limits and convergence.

Without the field structure, we could not reliably perform basic algebra. Without the order structure, we could not formulate inequalities, optimal choices, or comparative statics. Without completeness, we could not safely use limits, continuity, differentiation, integration, or asymptotic arguments.

Therefore, the concept of a field is the first layer of the foundation. It ensures that arithmetic works. The concept of an ordered field adds the ability to compare magnitudes. Finally, the completeness property ensures that the number system has no gaps. From this perspective, the real numbers are not just a familiar collection of numbers. They are a carefully structured mathematical system that supports the main tools used throughout modern economics, finance, statistics, and data science.

2 (Chinese version) 為什麼討論實數完備性之前要先介紹 Field？

在學習實數的完備性之前，許多數學分析教材會先介紹一個看起來比較抽象的概念：*field*，中文通常翻作「體」。對剛開始接觸高等數學的商管學院學生來說，這個概念一開始可能會顯得有些形式化，因爲教材會列出一長串公理，例如加法交換律、乘法交換律、結合律、分配律、加法反元素、乘法反元素等等。不過，field 的核心精神其實非常簡單：它描述的是一個可以穩定進行四則運算的數字系統。

我們先列出教科書中常見的正式定義，讓讀者知道 field 的嚴謹數學形式。[2]

Definition 2 (Field) *A set F is called a* field *if it has two operations defined on it, addition $x + y$ and multiplication xy, and if it satisfies the following axioms:*

(A1) If $x \in F$ and $y \in F$, then $x + y \in F$.

(A2) Commutativity of addition:

$$x + y = y + x$$

for all $x, y \in F$.

(A3) Associativity of addition:

$$(x + y) + z = x + (y + z)$$

for all $x, y, z \in F$.

(A4) There exists an element $0 \in F$ such that

$$0 + x = x$$

[2] See Jiří Lebl, *Basic Analysis I: Introduction to Real Analysis, Volume I*, Section 1.1.

for all $x \in F$.

(A5) For every element $x \in F$*, there exists an element* $-x \in F$ *such that*

$$x + (-x) = 0.$$

(M1) If $x \in F$ *and* $y \in F$*, then* $xy \in F$.

(M2) Commutativity of multiplication:

$$xy = yx$$

for all $x, y \in F$.

(M3) Associativity of multiplication:

$$(xy)z = x(yz)$$

for all $x, y, z \in F$.

(M4) There exists an element $1 \in F$*, with* $1 \neq 0$*, such that*

$$1x = x$$

for all $x \in F$.

(M5) For every $x \in F$ *such that* $x \neq 0$*, there exists an element* $1/x \in F$ *such that*

$$x(1/x) = 1.$$

(D) Distributive law:

$$x(y + z) = xy + xz$$

for all $x, y, z \in F$.

雖然這個定義看起來很長，但它的主要訊息其實很直觀。Field 就是在說：在這個集合裡，加法、減法、乘法與除法都能夠像我們熟悉的數字運算一樣正常進行，而且運算結果仍然留在同一個集合裡。換句話說，field 是一個可以「安全做代數運算」的數字系統。

例如，有理數集合

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z},\ b \neq 0 \right\}$$

就是一個 field。若我們取兩個有理數 $\frac{1}{2}$ 與 $\frac{1}{3}$，則

$$\frac{1}{2} + \frac{1}{3} = \frac{5}{6},$$

而 $\frac{5}{6}$ 仍然是有理數。同樣地，

$$\frac{1}{2} \times \frac{3}{4} = \frac{3}{8},$$

而 $\frac{3}{8}$ 也仍然是有理數。甚至只要分母不爲零，有理數彼此相除之後，結果仍然是有理數。例如

$$\frac{1}{2} \div \frac{3}{4} = \frac{1}{2} \times \frac{4}{3} = \frac{2}{3}.$$

因此，有理數在加、減、乘、除這些基本代數運算下是封閉的。這正是 field 的核心精神。

這點和整數集合

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$$

很不同。整數可以做加法、減法與乘法，例如

$$2 + 3 = 5 \in \mathbb{Z},$$

以及

$$2 \times 3 = 6 \in \mathbb{Z}.$$

但是整數不適合一般性的除法。例如

$$1 \div 2 = \frac{1}{2},$$

可是

$$\frac{1}{2} \notin \mathbb{Z}.$$

因此，整數集合 $\mathbb{Z}$ 不是 field。

更精確地說，整數缺少的是 field 定義中的乘法反元素，也就是 axiom (M5)。Field 要求每一個非零元素 x 都必須有一個倒數 $1/x$，使得

$$x \cdot \frac{1}{x} = 1.$$

在有理數 $\mathbb{Q}$ 中，2 的乘法反元素是 $\frac{1}{2}$，因爲

$$2 \cdot \frac{1}{2} = 1.$$

但是在整數 $\mathbb{Z}$ 中，$\frac{1}{2}$ 並不屬於整數。因此，2 在 $\mathbb{Z}$ 裡沒有乘法反元素。這就是爲什麼 $\mathbb{Z}$ 不是 field。

從商管與財務的角度來看，field 的重要性其實非常直觀。財務中最基本的折現公式是

$$PV = \frac{CF}{1+r},$$

其中 PV 是現值，CF 是未來現金流，r 是折現率。這個公式看起來只是簡單的財務數學，但它背後其實已經使用了 field 的概念。首先，$1+r$ 是加法運算；接著，將 CF 除以 $1+r$，其實等同於乘上 $1+r$ 的乘法反元素：

$$PV = CF \cdot \frac{1}{1+r}.$$

因此，只要我們要做折現、估值、報酬率計算、投資組合分析、效用最大化或最適化問題，就必須在一個能夠穩定處理加、減、乘、除的數字系統裡工作。這也是爲什麼 field 是經濟數學與財務數學的基礎結構。

不過，field 只處理代數運算的問題。它告訴我們加、減、乘、除是否可以正常進行，但它還沒有處理大小比較與極限的問題。因此，在實數理論中，field 通常只是第一步。下一步會引入 *ordered field*，也就是「有序體」。

有序體除了可以做四則運算之外，還可以比較大小。例如對任意兩個數 x 與 y，我們可以討論

$$x < y, \qquad x = y, \qquad \text{或} \qquad x > y.$$

有了順序之後，我們才可以談上界、下界、最大值、最小值、遞增數列、遞減數列等等概念。這些概念是討論極限與完備性的前提。

有理數 $\mathbb{Q}$ 不只是 field，它也是 ordered field。也就是說，有理數不但可以做四則運算，也可以比較大小。例如

$$\frac{1}{2} < \frac{2}{3}.$$

因此，若只看代數運算與大小比較，$\mathbb{Q}$ 似乎已經是一個很好的數字系統。可是，有理數仍然有一個根本問題：它不是完備的。

所謂不完備，直覺上可以理解爲：有理數數線上存在一些「洞」。最經典的例子是 $\sqrt{2}$。我們知道

$$\sqrt{2} \approx 1.41421356\ldots,$$

所以 $\sqrt{2}$ 位在 1 和 2 之間。但是

$$\sqrt{2} \notin \mathbb{Q}.$$

也就是說，雖然有理數非常密集，任意兩個有理數之間都還有其他有理數，但有理數仍然沒有填滿整條數線。某些數，例如 $\sqrt{2}$，就是不在有理數系統裡。

這件事可以透過數列來理解。考慮以下逐步逼近 $\sqrt{2}$ 的數列：

$$1, \quad 1.4, \quad 1.41, \quad 1.414, \quad 1.4142, \quad \ldots$$

這個數列的每一項都可以寫成有限小數，因此每一項都是有理數。從直覺上看，這個數列越來越接近某個固定的數，而那個數就是 $\sqrt{2}$。但是 $\sqrt{2}$ 不是有理數。因此，如果我們只在 $\mathbb{Q}$ 裡工作，就會遇到一個尷尬的情況：一個由有理數構成的數列明明越來越接近某個值，但那個值卻不在有理數系統裡。

這就是有理數不完備的意思。

實數 $\mathbb{R}$ 的完備性正是爲了解決這個問題。實數不只是 field，也不只是 ordered field；它是 *complete ordered field*。直觀地說，完備性表示實數數線沒有洞。只要一個數列應該收斂，它的極限就會存在於實數系統之中。

在數學分析中，完備性常常透過「最小上界性質」來表達。也就是說，如果一個非空集合有上界，那麼它一定存在一個最小上界。這個最小上界稱爲 supremum，記作

$$\sup A.$$

例如考慮集合

$$A = \{x \in \mathbb{Q} : x^2 < 2\}.$$

這個集合包含所有平方小於 2 的有理數。它有上界，例如 2 就是一個上界。但是在有理數系統 $\mathbb{Q}$ 裡，這個集合沒有最小上界。眞正應該作爲最小上界的數是

$$\sqrt{2}.$$

可是

$$\sqrt{2} \notin \mathbb{Q}.$$

因此，$\mathbb{Q}$ 缺少這個最小上界。這就是有理數不完備的具體表現。

相反地，在實數系統 $\mathbb{R}$ 裡，$\sqrt{2}$ 是存在的。因此，同樣的集合在 $\mathbb{R}$ 裡就有最小上界。這正是 $\mathbb{R}$ 與 $\mathbb{Q}$ 的本質差異。兩者都可以做四則運算，也都可以比較大小，但只有 $\mathbb{R}$ 是完備的。

因此，教材在討論實數完備性之前先介紹 field，是爲了建立一條清楚的邏輯路線：

$$\text{Field} \quad \Rightarrow \quad \text{Ordered Field} \quad \Rightarrow \quad \text{Complete Ordered Field}.$$

Field 探討的是：我們是否可以正常做加、減、乘、除？Ordered field 探討的是：我們是否可以比較大小？Complete ordered field 探討的是：當我們進行極限與逼近時，極限值是否仍然存在於這個數系中？

在這個層次架構下，有理數 $\mathbb{Q}$ 是 field，也是 ordered field，但它不是 complete ordered field。實數 $\mathbb{R}$ 則是 complete ordered field。因此，實數才是微積分、實分析、最佳化、機率論、計量經濟學與財務數學最自然的基礎數系。

經濟與財務模型中常見的變數，例如價格、報酬率、利率、消費、投資、效用、風險、機率與估計量，通常都被放在實數系統中分析，這是因爲實數具有足夠好的數學性質。若沒有 field 的代數結構，我們無法穩定地做基本運算；若沒有順序結構，我們無法比較大小與定義最佳化；若沒有完備性，我們就無法保證極限、連續性、微分、積分與最適解的存在。

因此，field 是理解實數完備性的第一層地基。它先確保四則運算可以正常進行；ordered field 進一步加入大小比較；complete ordered field 則確保數線沒有洞。從這個角度看，實數 $\mathbb{R}$ 不只是我們熟悉的數字集合，而是一個具有代數結構、順序結構與完備性的數學系統。正是因爲這些性質，實數才能支撐現代經濟學、財務金融、統計學與資料科學中大量使用的極限、最佳化與模型推導。